

Lab 01:Prepare hardware, software & network specification for basic computer

Scenario 1:

Hari works in the Media Department. His work involves in producing videos and images e.g. posters, web graphics, short videos and animations film making company. He wants you to produce a system design specification for his new multimedia computer system which he will use to run Macromedia Suite, Adobe Photoshop, Adobe Premier Pro and Microsoft Office. He would like high tech graphics and sound. The computer system must also be able to store large amounts of images in the company server and run quickly as he requires more than one program at a time and often needs to switch between them. In a report explain the choice of components briefly relating them back to the user requirements. It is important that final specification includes all the necessary hardware and software components for the system to work.

System Specification :-

Component	Company/Brand	Specification	Estimated Price (NPR)
Processor (CPU)	Intel	Core i9-13900K, 24-core	85,000
Graphics Card (GPU)	NVIDIA	GeForce RTX 4080 16GB	190,000
RAM	Corsair	32GB DDR5 6000MHz	25,000
Storage (SSD)	Samsung	1TB NVMe M.2 SSD	14,000
Secondary Storage	Seagate	4TB External HDD	12,000
Motherboard	ASUS	ROG Strix Z790-E Gaming WiFi	55,000
Power Supply	Cooler Master	850W Gold Certified	14,000
Casing	NZXT	Mid Tower ATX with RGB	18,000
Monitor	Dell	27" 4K UltraSharp	65,000
Sound Card	Creative	Sound BlasterX AE-5	20,000
Speakers	Logitech	Z625 THX Certified	18,000
Operating System	Microsoft	Windows 11 Pro	22,000
Software	Adobe	Creative Cloud (Photoshop, Premiere Pro)	50,000/year
Office Suite	Microsoft	Microsoft Office 2021	15,000

Networking	TP-Link	Gigabit LAN Card & Wi-Fi Adapter	5,000
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Conclusion

The above configuration ensures that Hari can run all his multimedia applications efficiently, multitask without lags, and store large files securely. The use of high-end components, especially the GPU, CPU, and RAM, caters to high-tech graphics and video processing demands. Additionally, the system includes reliable networking components for quick server communication and data transfer.

Scenario 2:

Sarah is a software developer in the Technology Department. Her work involves writing and testing code, running virtual machines, developing web applications, and managing databases. She also frequently uses design tools to create UI/UX prototypes. Sarah needs a powerful computer system that can handle heavy coding tasks, multiple virtual environments, and graphic design tools without any lag. She also requires high-speed internet connectivity and secure storage solutions for her projects. Create a system design specification for her new development workstation. The system should include high-performance components that support multitasking, fast data processing, and secure storage. In a report, explain the choice of components briefly, relating them back to the user requirements. Ensure that the final specification includes all the necessary hardware and software components for the system to work efficiently.

System Specification:-

Component	Company/Brand	Specification	Estimated Price (NPR)
Processor (CPU)	AMD	Ryzen 9 7950X 16-core	92,000
Graphics Card (GPU)	NVIDIA	RTX 4070 12GB	135,000
RAM	G.Skill	64GB DDR5 6000MHz	42,000
Primary Storage (SSD)	Western Digital	2TB NVMe M.2 SSD	22,000
Secondary Storage	seagate	4TB External HDD	12,000
Motherboard	MSI	MEG X670E ACE	68,000
Power Supply	Corsair	850W Gold Modular	18,000
Casing	Fractal Design	Meshify 2 Mid Tower	20,000
Monitor	LG	32" 4K UHD Monitor	70,000
Keyboard	Keychron	K8 Mechanical Wireless	15,000
Mouse	Logitech	MX Master 3	13,000
Operating System	Microsoft	Windows 11 Pro	22,000
Development Tools	JetBrains	All Products Pack (Yearly)	20,000
Design Software	Figma	Professional Plan (Yearly)	10,000

Office Suite	Microsoft	Microsoft Office 2021	15,000
Networking	TP-Link	Wi-Fi 6E PCIe Card + Gigabit LAN	6,000
Security	Bitdefender	Total Security (1 Year)	5,000

Conclusion

This workstation configuration ensures that Sarah can develop, test, and deploy software applications efficiently. The powerful CPU and large RAM capacity support multiple virtual machines and resource-intensive tools. The SSD offers fast boot times and data access, while the GPU enables design and UI/UX rendering. Reliable networking and security software ensure safe and high-speed connectivity, making this system ideal for a modern software developer.